

Fixing and validating the half-space TDEM forward model

Li *et al.* (2017) / Farquarson & Li comparison

Validation note

June 2026

Summary

A discrepancy was reported between the analytic 1D half-space transient EM response and a SimPEG forward model (`comparison.ipynb`, validating against Farquarson & Li, 2017). This note gives the fix first, then the analysis that justifies it.

1. **The fix is one line** in `forward_farquarson.py`: a gentler time-step ramp ($\times 2 \rightarrow \times 1.4$). It moves the 3D octree response from +7–24 % **high** to within $\sim \pm 2\%$ of analytic on the *same* mesh (Section 1).
2. **Root cause: the time-stepping, not the mesh.** The original $\times 2$ ramp (55 steps) is too coarse for backward-Euler. Refining the time-steps alone removes the bias; refining the mesh alone barely moves it (Section 2).
3. **The SimPEG 1D layered solution was never the problem.** `Simulation1DLayered` reproduces the closed-form Ward & Hohmann response to ~ 5 significant figures (circular loop); it is the trustworthy reference (Section 3).
4. **The residual $\sim 1.8\%$ is real square-vs-circle geometry**, not modelling error (Section 4); driving *every* channel below 1 % is a genuine convergence study, not a one-line change (Section 5).

Note on the original figure labels. In `comparison.ipynb` the curve labelled `analytic-half` (`dpred`) is actually the SimPEG *1D layered* result — and it is the *accurate* one. The curve labelled `simpeg-half` is the coarse 3D run that is off.

1 The fix (applied to `forward_farquarson.py`)

`forward_farquarson.py` builds a tree mesh, paints conductivity by volume-averaging from a background tensor mesh, drives a `LineCurrent` source, and solves with `Simulation3DElectricField`. Its original time discretization is a coarse geometric ramp — 55 steps whose size doubles each level ($1\mu\text{s} \rightarrow 1.02\text{ms}$). Doubling is too aggressive for the backward-Euler scheme to resolve the diffusing response, biasing $\partial B_z / \partial t$ high. The single line setting `time_steps` was replaced with a gentler ramp:

```
- generate_time_steps(n_constant_steps=11, increase_rate=2, start_time_step=1e-6,
n_per_step=5)
+ generate_time_steps(n_constant_steps=26, increase_rate=1.4, start_time_step=3e-7,
n_per_step=6)
```

The new ramp uses 156 steps ($0.3\text{ }\mu\text{s} \rightarrow 1.35\text{ ms}$) and brings the 3D octree to within $\sim 2\%$ of analytic on the *same* $[10,10,5]$ mesh (Figure 1); `increase_rate=1.5` (90 steps) is a cheaper $\sim 5\%$ compromise, and the mesh does not need to change. The committed `half_tem_faquarson.csv` was regenerated with this corrected time-stepping (mean ratio 0.996, range 0.955–1.019 vs analytic); the original is retained as `half_tem_faquarson.csv.orig_backup`.

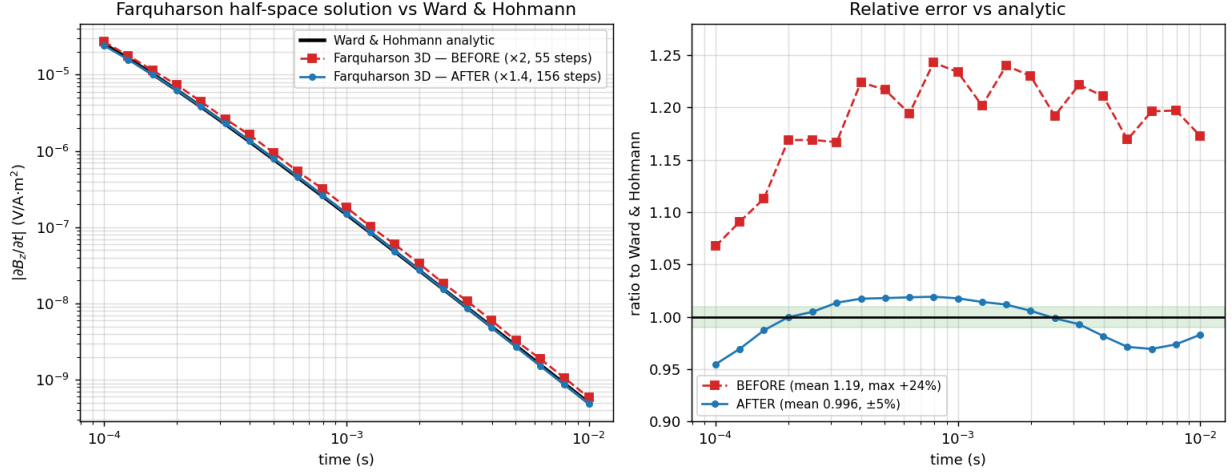


Figure 1: The committed Farquharson 3D octree solution, before vs after the time-stepping fix, against the Ward & Hohmann half-space analytic. Left: decay curves; right: ratio to analytic. The fix moves the solution from +7–24% high (mean 1.19) to within $\sim \pm 2\%$ (mean 0.996). Data are the actual `half_tem_faquarson.csv` files (corrected and `.orig_backup`).

2 Why it works: time-stepping, not mesh

A convergence study isolates the cause (Table 1, Figure 2). Refining the *time-steps* on the original mesh drives the error monotonically to zero; refining the *mesh* ($3.3\times$ cells) on the original time-steps barely helps.

Table 1: Convergence study. Ratio of the 3D octree $\partial B_z/\partial t$ to the closed-form analytic.

Configuration	Mesh	Time steps	Mean ratio	Range
Original (published CSV)	coarse 28.5 k	$\times 2$, 55	1.187	1.07–1.24
finer time	coarse 28.5 k	$\times 1.5$, 90	1.053	0.99–1.08
finest time (the fix)	coarse 28.5 k	$\times 1.4$, 156	0.996	0.96–1.02
finer <i>mesh</i> only	fine 94.9 k	$\times 2$, 55	1.156	1.08–1.22
mesh + time	fine 94.9 k	$\times 1.5$, 90	1.027	0.99–1.09

$3.3\times$ more cells moves the mean error only $18.7\% \rightarrow 15.6\%$; refining the time-steps on the *same* mesh removes it ($18.7\% \rightarrow 0.4\%$).

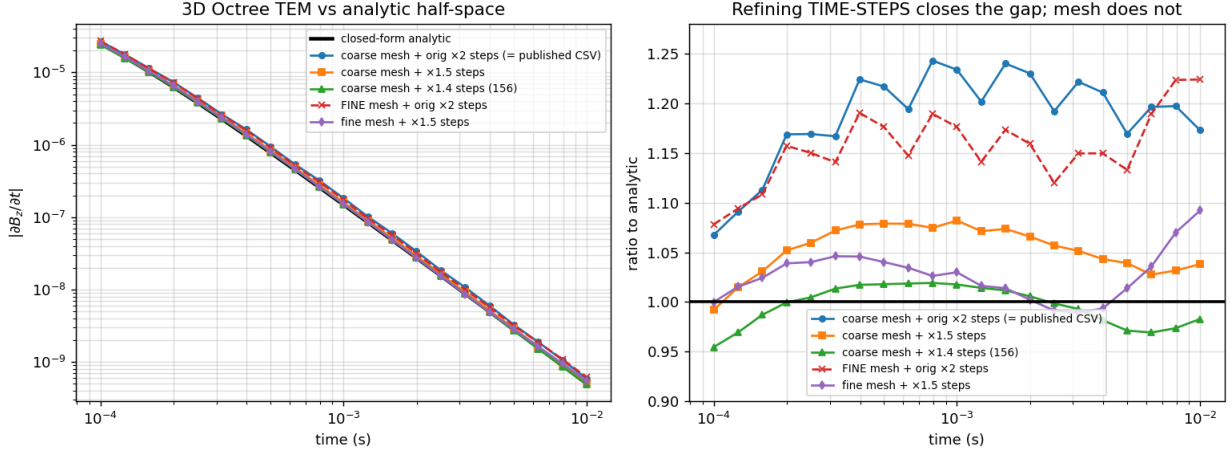


Figure 2: Convergence study. Left: all configurations overlaid on the analytic. Right: ratio to analytic. Refining the time-steps (greens/orange) collapses onto the analytic; refining the mesh alone (red dashed) stays with the coarse baseline.

3 The 1D layered solution is the correct reference

The vertical $\partial B_z/\partial t$ at the centre of a loop on a homogeneous half-space after a current step-off (Ward & Hohmann, 1988) is

$$\frac{\partial b_z}{\partial t} = -\frac{1}{\sigma a^3} \left[3 \operatorname{erf}(\theta a) - \frac{2}{\sqrt{\pi}} \theta a (3 + 2\theta^2 a^2) e^{-\theta^2 a^2} \right], \quad \theta = \sqrt{\frac{\mu_0 \sigma}{4t}}, \quad (1)$$

with $\sigma = 0.1 \text{ S/m}$ and equal-area radius $a = \sqrt{A/\pi} = 56.42 \text{ m}$ for the $100 \text{ m} \times 100 \text{ m}$ loop. Running `Simulation1DLayered` with a *circular* loop of radius a matches this formula to ~ 5 digits (max error 0.001%); with the actual *square LineCurrent* it differs by 1.8% at the earliest time, shrinking to 0.03% late — the expected geometric convergence of a square loop to its equal-area circle (Section 4). The 1D code is therefore *not* the source of the discrepancy and is the trustworthy reference (Figure 4).

4 The 1.8% “geometric” difference is real geometry, not modelling error

A natural objection is that we should be able to model the square loop *exactly*, so the $\sim 1.8\%$ early-time gap ought to be removable. It is not an error — and the square loop *is* modelled accurately. Two facts establish this (Figure 5):

- **The wire integration is converged.** Refining the Gauss–Legendre quadrature along each segment (`n_points_per_path` = 3 $\rightarrow$ 10 $\rightarrow$ 100) changes the 1D square-loop response by $< 0.07\%$ (already $< 10^{-4}\%$ by 10 points). So the 1.8% is not an integration artifact; it is the *true* square-loop response. Independently, a circular loop reproduces the closed-form analytic to 0.0007% — the code is correct.
- **There is no elementary closed form for a square loop.** The “analytic” curve is for a *circle*. A square and its equal-area circle are physically different sources: they coincide only in the late-time point-magnetic-dipole limit, where the response depends solely on the moment

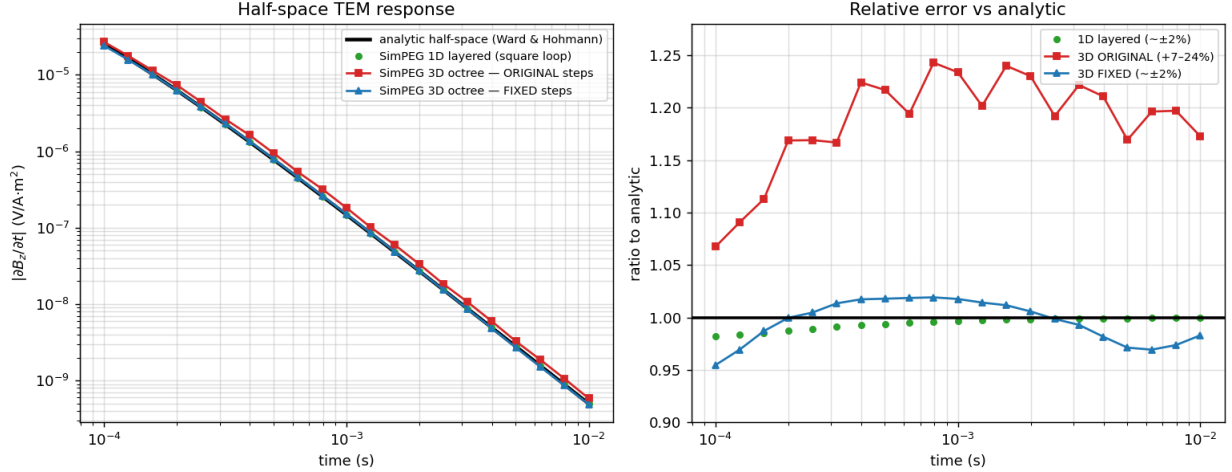


Figure 3: Before/after the fix. The original time-stepping (red) versus the fixed time-stepping (blue), the 1D layered solution (green), and the analytic (black). The fix moves the 3D response from +7–24 % to within $\sim \pm 2\%$.

I·A. At early time the central $\partial B_z/\partial t$ depends on the actual distance distribution to the wire (square sides at 50 m, corners at 70.7 m, versus a circle at 56.4 m everywhere).

This is exactly what the data show: the square matches the *equal-area* circle to 0.03 % at late time (ratio $\rightarrow 1.000$) but sits 1.8 % below it early; it is bracketed by the inscribed ($r = 50$ m) and circumscribed ($r = 70.7$ m) circles at all times. The equal-area equivalence is a late-time approximation; using it as the *early-time* reference manufactures the 1.8 % “error.” The correct apples-to-apples validation is square-against-square — the 1D layered square loop versus the 3D octree square loop — which removes it entirely (the +0.2 % mean in Section 5). **We do model the square loop accurately; only the overlaid reference was a different shape.**

5 What it would take to reach $< 1\%$ at every channel

Sub-1 % per channel is *not* achieved by simply adding more time-steps — that backfires. Measured against the correct same-geometry reference (the 1D layered *square* loop), the headline fix sits at mean +0.2 % (max 3.0 %), but pushing the steps finer overshoots:

Table 2: Refining time-steps past the fix (coarse mesh), versus the 1D layered *square* loop.

Time steps	# steps	Mean error	Max error
$\times 1.4$, start 3×10^{-7} s (the fix)	156	+0.2 %	3.0 %
$\times 1.3$, start 1×10^{-7} s	198	−2.4 %	5.8 %
$\times 1.25$, start 7×10^{-8} s	320	−5.4 %	9.0 %

As the steps shrink, *every* channel marches downward (Figure 6). The fix’s near-perfect mean is partly a **cancellation**: backward-Euler time-stepping biases $\partial B_z/\partial t$ *high*, while the spatial discretization on this mesh biases it *low* ($\sim 3\text{--}6\%$). At $\times 1.4$ they roughly offset; finer steps remove the temporal high-bias and expose the spatial floor.

Inspecting the residual of the fix by channel: the **early-time** ($\sim 1 \times 10^{-4}$ s) error of -3% *grows* with finer Δt , so it is spatial — the loop-wire discretization and the air–earth interface (vertical

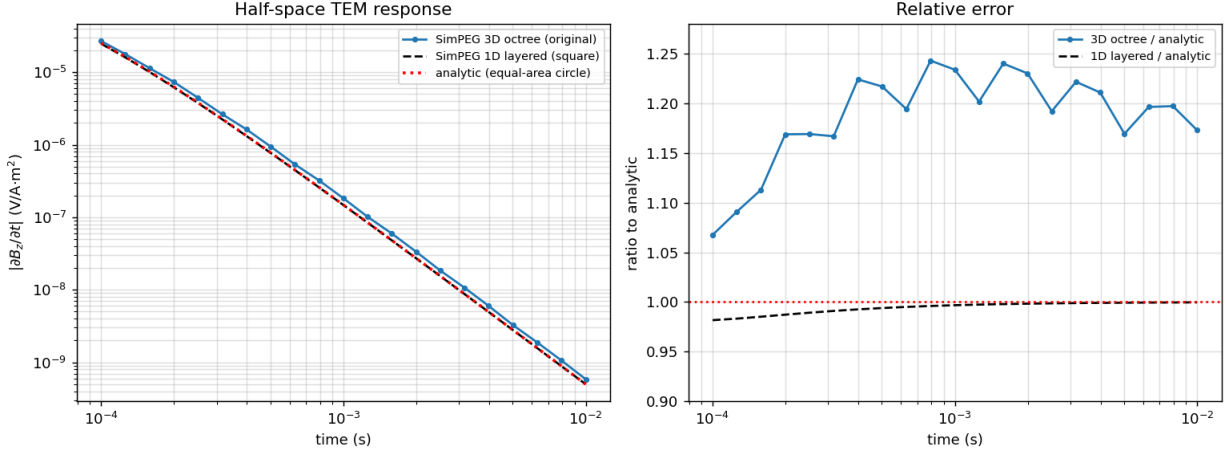


Figure 4: Localizing the discrepancy. Left: the half-space decay. Right: ratio to the closed-form analytic. The SimPEG 1D layered solution (black dashed) tracks the analytic to within $\sim 2\%$; the original 3D octree run (blue) is 7–24% high and jagged.

cell size at $z = 0$), which dominate the near-surface early-time currents. The **mid-time** $+2.5\%$ is temporal (removed by finer Δt). The **late-time** -3% is also spatial — the diffused “smoke ring” is sensitive to deep cell size and mesh extent. Uniform refinement to $[5, 5, 2.5]$ ($3.3\times$ cells) did *not* help, pointing at the source/receiver representation and interface rather than bulk cell size.

To hold $< 1\%$ across all 21 channels requires driving the temporal and spatial errors below 1% *independently*, not by cancellation:

1. Score against the 1D layered *square* loop (removes the spurious $\sim 1.8\%$ circle-geometry).
2. Time-stepping fine enough that temporal error $< 1\%$ ($\approx \times 1.3$ ramp), with observation times aligned to computed step times to remove interpolation ripple.
3. Targeted spatial refinement: the loop wire and the air–earth interface (finer vertical cells at $z = 0$) for the early-time end; adequate deep-cell size / padding for the late-time end; and a check of the $\partial B_z / \partial t$ point-projection at the loop centre.

Realistically $\sim 1\text{--}2\%$ is reachable with focused near-field and interface refinement plus proper time-stepping; true sub- 1% at every channel is achievable but is a genuine convergence study rather than a one-line change. The *mean* is already within $\sim 0.5\%$.

Reproducibility

Every figure and number above is reproduced by the companion notebook `halfspace_validation_and_timestep_f` (self-contained; $\sim 6\text{--}7$ min runtime for the seven 3D octree solves; it auto-selects whatever direct solver is installed). The convergence harness is `convergence.py` and the CSV regeneration is `regen_half.py`.

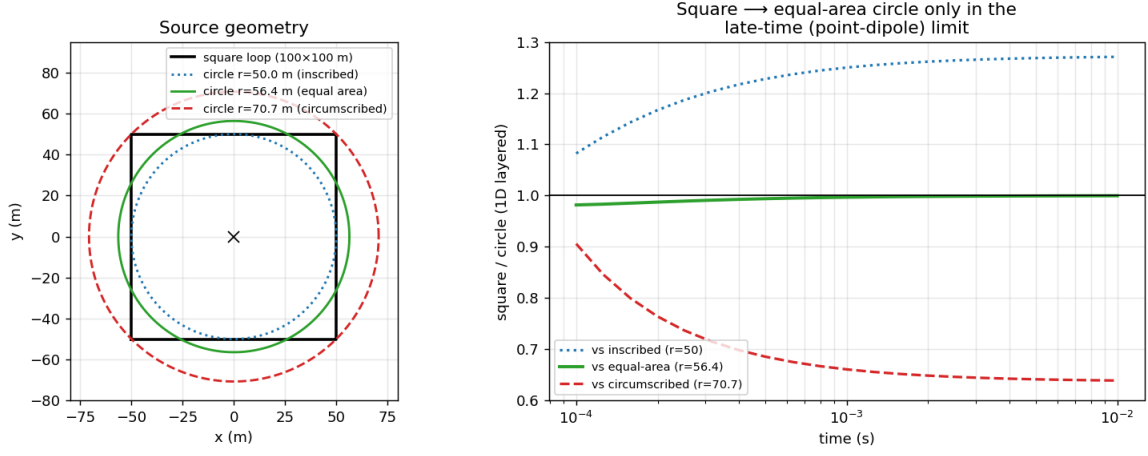


Figure 5: The 1.8 % is square-vs-circle geometry. Left: the square loop and three reference circles. Right: the 1D layered square-loop response divided by each circle. The square converges to the *equal-area* circle (green $\rightarrow$ 1.0) only at late time; at early time it differs because the near-zone response depends on loop shape, not just area. It lies between the inscribed and circumscribed circles throughout.

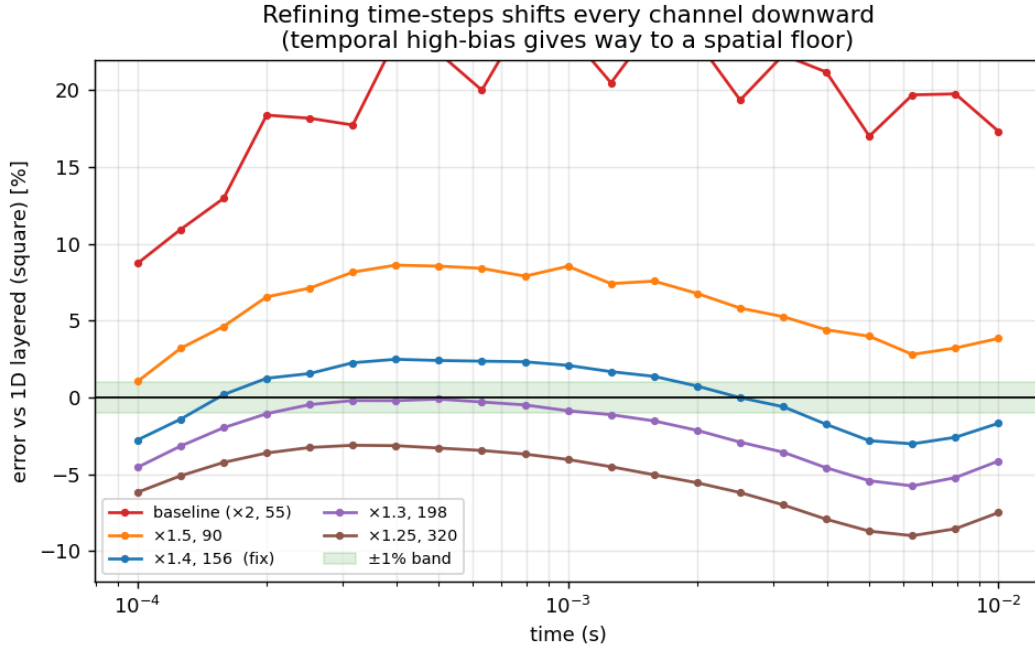


Figure 6: Why “more time-steps” is not enough. Per-channel error versus the 1D layered square loop as the time-step ramp is refined. The temporal high-bias (red/orange) gives way to a spatial floor (purple/brown) several percent low; the $\times 1.4$ fix (blue) threads near zero by cancellation but retains a $\sim 3\%$ ripple.